Harshit Singh

Machine Learning Engineer | Python • PyTorch/TensorFlow • Computer Vision • MLOps

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Education

Vellore Institute of Technology, Chennai — Chennai, India
B.Tech, Electronics and Computer Engineering — CGPA: ***7.97*** — *2023 – 2027*

New Era School (CBSE) — Ghaziabad, Uttar Pradesh
Class XII: ***87.16%*** *— Class X:* ***89.83%*** — *2020 – 2023*

Experience

AI & Robotics Intern — Autonomous Mobile Robotics [**Certificate**] [**GitHub**] — Jul 2025 – Aug 2025
STEMTEC AI & Robotics Technology Pvt. Ltd. — *Chennai, India*

- Engineered a **15 Hz** Extended Kalman Filter sensor-fusion pipeline (ROS 2 `robot_localization`) fusing wheel-encoder odometry with 6-DOF IMU data, cutting mean spatial drift by **60%** (**0.45 m → 0.18 m**) and angular error by **69.2%** across 10 m ground-truth trials.
- Built a real-time reactive perception pipeline processing **∼3,600 points/sec** of LiDAR telemetry at **10 Hz** with windowed sector-filtering, achieving **95% obstacle-detection accuracy** across 20 randomized stress-test trials at **sub-50 ms** end-to-end latency.

Projects

SPARC — Edge ML & Multi-Modal Perception Engine *(PyTorch, YOLOv8, TensorFlow.js)* [**Demo**] [**Web-App**]
- Engineered a low-latency edge ML pipeline (<**8ms/sample**, **30 FPS**) on ARM Cortex-A72, cutting CPU compute **35%** via 229-dim landmark vectorization (**99.97%** compression).
- Orchestrated multi-model inference (YOLOv8 Nano + CNN) with a 45-frame sliding-window buffer, cutting false-positives **94%** — **Top 5 finalist** (500+ teams), SIH 2025. [**Certificate**]

Weather Risk AI — Time-Series Forecasting & Risk Modeling *(Keras/TensorFlow, NetworkX)* [**GitHub**]
- Architected a 2-layer stacked LSTM (Keras/TensorFlow) achieving **sub-0.05 MSE** across 6 features, via a zero-leakage **(N,10,6)** sliding-window pipeline with serialized `MinMaxScaler` artifacts.
- Built graph-based risk solvers (NetworkX, Edmonds-Karp max-flow, Dijkstra) modeling 5-step diffusion across an 8-node network and a **10×10** pressure grid.

DrishtiGuide — Edge CV & Assistive Perception System *(PyTorch, OpenCV, YOLOv8)* [**GitHub**]
- Deployed a real-time YOLOv8n pipeline (buffer flushing, 3-frame subsampling), cutting compute overhead **66.7%** at **sub-100ms** end-to-end latency.
- Built an automated LBPH retraining pipeline (zero manual cold-start) and a 6-DOF IMU fall-detection classifier with monocular distance estimation (≤**3cm** error).

NeuroBridge AI — Multimodal Clinical ML Platform *(MediaPipe, TensorFlow.js, Node.js)* [**Live**] [**GitHub**]
- Architected an edge CV pipeline (MediaPipe FaceLandmarker, WebGL) extracting **468** 3D landmarks at **30 FPS** in-browser, feeding a multimodal late-fusion clinical risk model.
- Deployed and served a TensorFlow.js NLP toxicity classifier (τ=**0.85**, 7 categories) in production with resilient model serving (keep-alive + backoff), sustaining **99.9%** uptime.

Skills

Languages: Python, C, SQL, Java
ML/DL Frameworks: PyTorch, TensorFlow/Keras, OpenCV, Scikit-Learn
Computer Vision, NLP & Edge ML: YOLOv8, MediaPipe, NLP, Model Serialization, Edge Inference Optimization, Time-Series Forecasting
MLOps & Systems: AWS, Docker, CI/CD, Linux (Ubuntu), Bash, ROS 2, Git, REST APIs
Professional Skills: Data-Driven Problem Solving, Cross-Functional Collaboration, Communication, Time Management

Achievements & Leadership

- **3rd Place, Technovation 2025** (Kurukshetra '25, CEG Guindy, Anna University) — ranked **2nd Runner Up out of 50+** teams; **INR 15,000** award. [**Certificate**]
- **Winner, Innovate X Impact (V-NEST)** — secured **1st Position out of 35** teams; **INR 5,000** award.
- **Deputy Operations Lead, Team Namo Nirvana (VIT Chennai)** — led **100+ member** teams across **5–6 competitions/semester**, boosting efficiency by **35%** and securing **2 sponsors.** [**LOA**] [**Certificate**] [**LOR**]

Languages

English (Fluent) | **Hindi** (Native)